

Civics Group	Index Number	Name (use BLOCK LETTERS)
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H2

**ST. ANDREW'S JUNIOR COLLEGE
2017 JC2 PRELIMS**

H2 BIOLOGY**9744/4****Paper 4: Practical Exam**

Friday

25th August 2017

2 hours 30 minutes

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Shift
Lab

INFORMATION TO CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Section A	
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1	/ 21
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2	/19
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3	/15
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Total	/55
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IMPORTANT:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use it. Please answer **QUESTION 2(b)** within this time frame.

Candidates with no access to microscope at the start of the paper will be given access **1h 15min after the start of the paper**. You may proceed with **QUESTION 1** first.

QUESTION 1**Investigation into the effect of enzyme concentration on the hydrolysis of starch**

The enzyme amylase catalyses the hydrolysis of starch.

You are required to investigate the effect of the concentration of amylase on the time taken to completely hydrolyse starch.

Iodine solution turns from yellowish brown to blue-black when starch is present. The time taken for the complete hydrolysis of starch can be found by removing a sample of an amylase and starch mixture at regular time intervals, and adding it to a drop of iodine solution. The starch has been completely hydrolysed when the iodine solution remains yellowish brown after adding the sample. This is the end-point. In judging the end-point, specks of blue-black in an otherwise yellowish brown solution can be ignored.

You are provided with:

- 40.0 cm³ of 2.0% amylase solution, labelled **E**, which is an irritant,
- 100.0 cm³ of distilled water, labelled **W**,
- 40.0 cm³ of starch solution, labelled **S**,
- Iodine solution, labelled **iodine**, which is a stain.

Read steps 1 to 9 before starting.

Proceed as follows:

You are required to prepare different concentrations of the amylase solution and set up a control.

1 Carry out **simple** dilutions of the amylase solution, **E**, to obtain a range of concentrations in which the concentration of the amylase is reduced by 0.4% between each successive dilution.

Prepare 10.0 cm³ for each concentration of amylase solution, **using the plastic containers provided**.

Complete Table 1.1 to show how you will prepare the different concentrations of amylase solution.

.....[2]

Table 1.1

Concentration of amylase solution / %	volume of E / cm ³	volume of W / cm ³
2.0	10.0	0.0
1.6	8.0	2.0
1.2	6.0	4.0
0.8	4.0	6.0
0.4	2.0	8.0

a) Correct selection of [amylase]

b) Correct volumes of enzyme to make each concentration **and** correct volumes of water to make an appropriate total volume to 1dp e.g. 10.0 cm³ (must be the same volume for each concentration)

[Deduct one mark if no lines drawn within the table]

[Deduct one mark if volume precision is incorrect]

2 Prepare a suitable control for this investigation.

Describe the control that you have prepared. Explain the rationale of the control.

.....[2]

a) [Setup] Replace enzyme with equal volume / 2.0 cm³ of distilled water/W

b) [Rationale] To prove that the disappearance of blue-black coloration observed is due to the action of amylase enzyme on starch

You are required to investigate the effect of different amylase concentrations on the time taken to completely hydrolyse starch. You will test samples from mixtures of a starch solution and different concentrations of amylase solution at a chosen time interval until each end-point is reached, up to a maximum of **180 seconds**.

3 Decide on a suitable time interval at which samples from each mixture of amylase solution and starch solution will be tested for complete hydrolysis of starch.

State the chosen time interval with a reason for this choice.

Time interval

Reason

..... [1]

a) Select time interval of 30s or less

+ Reason: Ref. short time interval will give more precise results [Reject: more accurate results]
 / Ref. a not too large time interval such that accuracy is not compromised
 / to ensure that at least 6 readings are obtained during the 180s
 [Reject 1min due to a lack of readings]

4 The enzyme's optimum temperature is 35°C. Using the hot and tap water provided, set up a water-bath in the beaker labelled **water-bath**, so that you can maintain this temperature throughout the investigation.

5 Put uniform-sized drops of iodine solution on the white tile, labelled with the times that a sample from each mixture of amylase solution and starch solution will be removed and tested, as shown in Fig. 1.1.

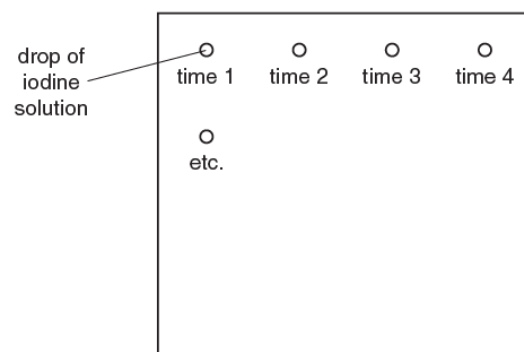


Fig. 1.1

6 Put 3.0 cm³ of **S** into a test-tube and 2.0 cm³ of 2.0% amylase solution into a separate test-tube. Put the test-tubes into the water-bath for at least one minute in order to equilibrate to 35°C

7 The reaction will start as soon as **S** and the amylase solution are mixed.

Add the starch solution, **S**, to the 2.0% amylase solution, and start timing immediately. Using a Pasteur pipette, remove a sample of the mixture at the first chosen time and add **one drop** to the first drop of iodine solution on the white tile. Continue removing and testing samples at the chosen time interval until the end-point is reached, up to a maximum of **180 seconds**. *Make sure that the mixture or enzyme and starch are maintained at 35°C throughout the experiment.*

8 Repeat steps **5–7** to collect the result for each of the other concentrations of amylase solution and the control that you have prepared. Record '**more than 180**' for any mixtures that have not reached the end-point by 180 seconds.

9 Use the space below to record your results.

..... [4]

Table showing time taken for different amylase concentration to completely hydrolyse starch / end-point

Concentration of amylase solution / %	Time taken for end-point / s
2.0	40
1.6	60
1.2	100
0.8	120
0.4	160
0.0	More than 180

Results are based on 20s time interval

- a) Table format: Independent variable in leftmost vertical column;
 b) Suitable column / row headings with correct units; [Reject: concentration of E as E means 2%; Reject: time **interval** when end point is reached]
 c) Records results for 5 concentrations of amylase **and control**; to **whole number**
 d) Trend of shortest time for highest concentration of amylase to longest time for lowest concentration of amylase, and control as "more than 180";
 [Penalise 1m for writing units beside every reading e.g. 40s, 60s etc]

[Penalise 1m for absence of boundary and/or grid lines; do not double penalise if Q1 is already penalised]

10 Explain how the concentration of amylase affects the rate of hydrolysis of starch.

.....[3]

- a) As the concentration of amylase increases, the rate of hydrolysis of starch increases;
 b) Increase in frequency of effective collisions between enzyme and substrate molecules ;
 c) Increase in concentration of enzyme-substrate complexes formed per unit time ;
 (Increase in concentration of products formed per unit time)

AVP:

- d) Ref. Increase in amylase concentration; increase in number of active sites

11 Temperature was one variable which was controlled in this investigation.

Identify one variable that affects enzyme reactions, which was not controlled in this investigation.

.....[1]

- a) pH

12 Suggest how you would control this variable.

.....[1]

a) Add the same volume of pH buffer to each test

[Reject: buffer as it is too vague; isotonic buffer; PBS buffer as it is isotonic buffer]

14 For a biotechnological process involving an enzyme to work most efficiently, the enzyme must work at its maximum rate, R.

An enzyme can be used to catalyse the conversion of ethanol (substrate) to acetaldehyde (product).

The effect of the concentration of ethanol (A) on the maximum rate of the production of acetaldehyde (R), is shown in Table 1.2.

Table 1.2

Concentration of ethanol (A) / mol dm ⁻³	Maximum rate (R) / min ⁻¹
0.00800	0.0700
0.0150	0.110
0.0500	0.170
0.100	0.220
0.300	0.270

A linear graph can be drawn by plotting 1/R against 1/A.

This can be used to find R (maximum rate of production) for any particular ethanol concentration in this range.

Complete Table 1.3 for the values of A and R in the last row of Table 1.2, by calculating 1/A and 1/R to the appropriate number of decimal places.

.....[1]

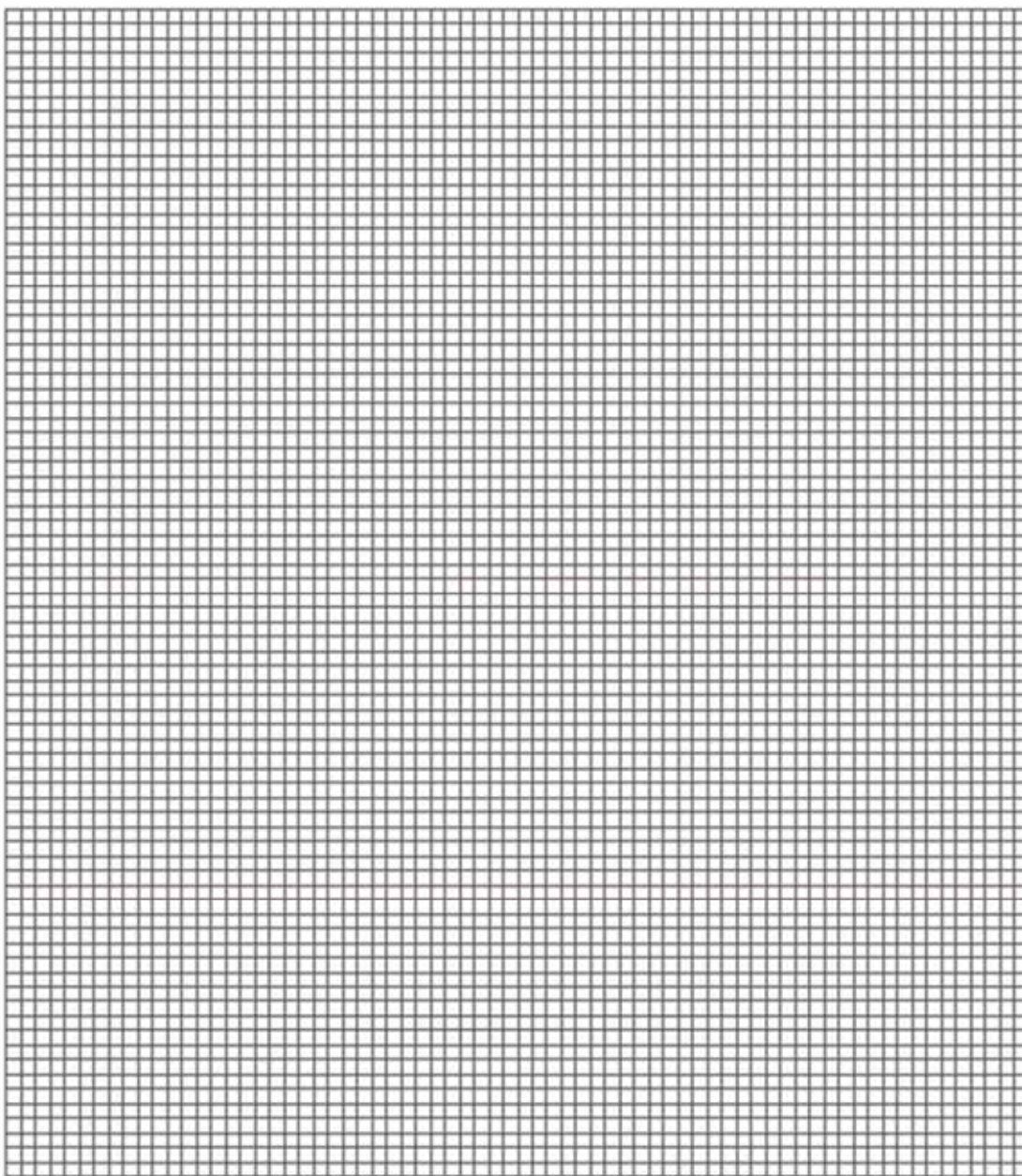
Table 1.3

1/A / mol ⁻¹ dm ³	1/R / min
125.0	14.3
66.7	9.1
20.0	5.9
10.0	4.6
3.3	3.7

a) Correct calculation of 1/A and 1/R, and rounding to 1 dp (follow the pattern in the prior rows)

15 Using the data from Table 1.3, draw a graph on the grid provided.

- [4]
- a) use of sensible scale [Reject: odd scales e.g. 3:10] that allow points to occupy at least half the grid in both x and y directions
 - b) all points plotted accurately to within half a small square on the grid
 - c) axes correctly labelled with correct units, ascending scale and equidistant intervals
 - d) correct straight line of best fit [Reject: extrapolation of points]; cut through 1 point and ensured that there are equal number of points on either side of the best fit line



16 Find the maximum rate of production (R) which would be achieved if the ethanol concentration (A) was 0.1 mol dm^{-3} .

Show clearly how you obtained R.

R=min⁻¹ [2]

- a) Calculation of $1/A = 1/0.1 = 10.0 \text{ mol dm}^{-3}$
- b) Correct reading from graph (precision to half a small square); correct conversion from $1/R$ to R and answer expressed to 3 sig fig; allow ECF from incorrect calculation of $1/A$

[TOTAL : 21]

QUESTION 2

Fig 2.1 is a photomicrograph of a stained transverse section through part of a leaf from a different type of plant.

You are not expected to be familiar with this specimen.

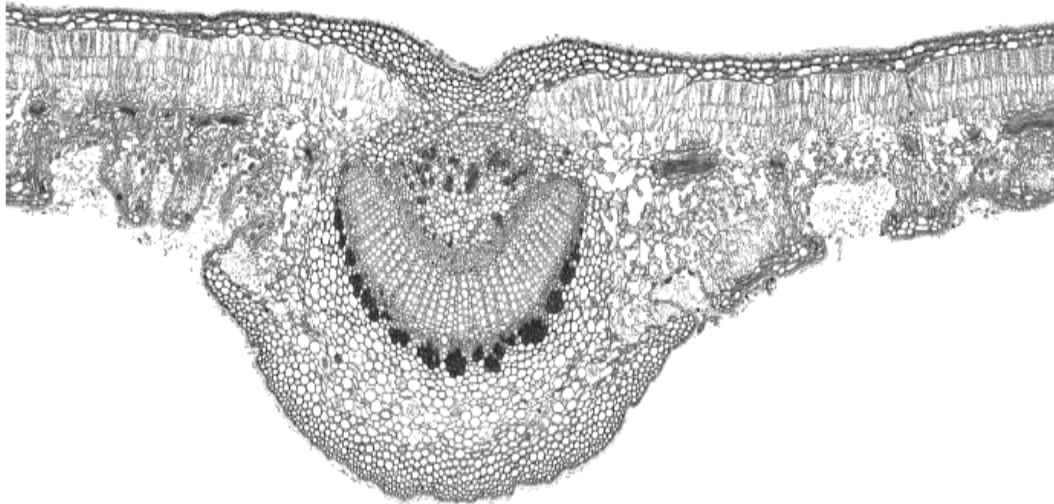


Fig. 2.1

(a) Draw a large **plan diagram** of Fig. 2.1 in the space provided below. Please refer to the coloured photo micrograph provided on the student's bench. [2]

Mark scheme:

- 1** at least 4 lines at leaf blade + size at least 60 mm thickness at mid-rib region + no shading ;
- 2** no cells drawn + at least 5 layers within mid-rib region

REMINDER:

Candidates with access to microscope at the start of the paper are given the **first 1h 15 min** to use it. Please answer **QUESTION 2(b)** within this time frame.

(b) You are required to measure the diameter of the field of view using the clear plastic ruler.

Proceed as follows:

1. Put the clear plastic ruler on the stage of the microscope and view the scale lines using low power ($\times 100$).
2. Measure the diameter of the field of view and record this in **(b)(i)**.

(i) Diameter of the field of viewmm [1]

a) Accept: 1.6 – 1.8 mm

Fig. 2.2 is the same photomicrograph as in Fig. 2.1 showing the field of view at the same magnification as the field of view you have just measured.

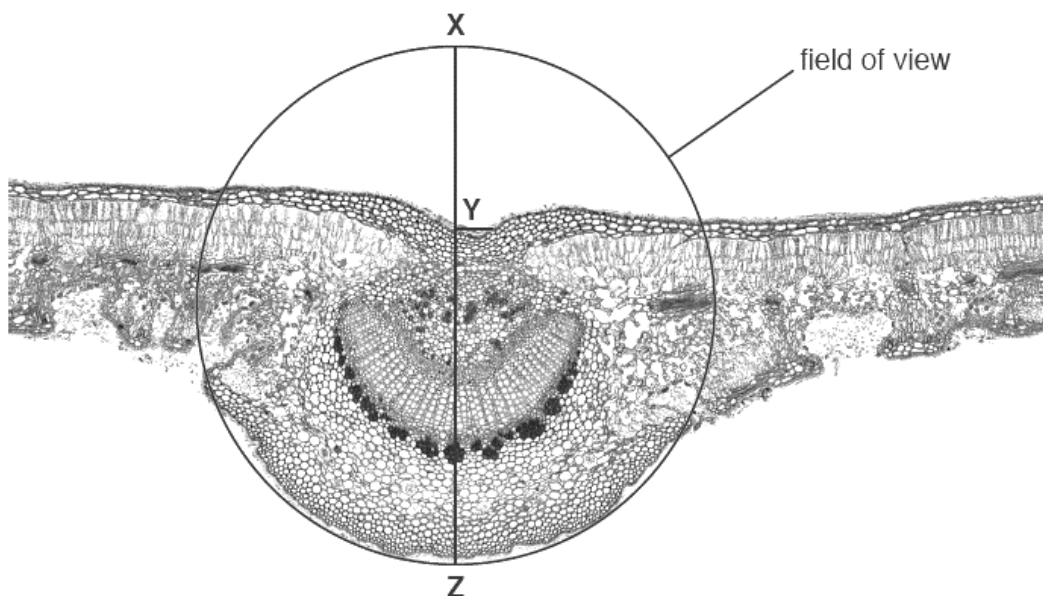


Fig. 2.2

(ii) Using appropriate measurements, calculate the fraction of the diameter of the field of view occupied by the leaf in Fig. 2.2 along the line **X–Z**.

fraction of diameter of field of view [1]

$$YZ = 4.4 - 4.5\text{cm}$$

$$XZ = 6.8 - 6.9\text{cm}$$

[Reject: answers in decimal places]

Fraction of diameter =

$$4.4 / 6.8 = 11 / 17$$

$$\text{or } 4.4 / 6.9 = 44 / 69$$

$$\text{or } 4.5 / 6.8 = 45 / 68$$

$$\text{or } 4.5 / 6.9 = 15 / 23$$

(iii) Using your answers to (b)(i) and (b)(ii) calculate the depth of the midrib, as shown by line Y–Z. Give your answer to the nearest μm .

You may lose marks if you do not show your working.

..... μm [2]

a) shows answer to (b)(i) multiplied by answer to (b)(ii)

b) decision to multiply by 1000 (to convert to μm)

(iv) A student used a clear plastic ruler to measure the field of view of a microscope. The student replaced the ruler with a slide of a leaf and **estimated** the diameter of the midrib. Using these results the student calculated the actual diameter of the midrib.

State how this student could have modified their method to obtain a more accurate result. State the apparatus the student would use and describe the method.

apparatus

.....

method

.....[3]

Apparatus :

- 1 stage micrometer **and** eyepiece graticule ;

Method:

- 2 Calibration of eyepiece graticule:

Place stage micrometer on the stage of the microscope and position stage scale such that it is **superimposed** on or aligned next to the smaller eyepiece scale under low power objective lens (x10).

Count the number of eyepiece unit to 1 stage unit.

Calculate the length of each eyepiece unit by dividing the length of 1 stage unit (e.g. 0.1mm) by the number of eyepiece units (e.g. 10).

/ Given that 1 unit of the stage scale measures 0.1 mm, the length of 1 eyepiece unit (division) can be measured by dividing the length of 1 stage unit by the number of eyepiece units that can be fitted on to it. E.g. 0.1 mm divide by 10 eyepiece unit = 0.01 mm = 10 μm

- 3 **Measurement** of midrib using eyepiece graticule by **counting the number of eyepiece units** which is then **multiplied** by the **length of each eyepiece unit** (e.g. 10 μm)

(c) One technique used for studying antigen-antibody reactions is immunodiffusion.

Wells are cut into an agar support medium to contain antigens and antibodies. Antibodies and antigens diffuse out of the wells into the agar. If an antigen meets a complementary antibody a reaction occurs causing a band of precipitate to appear.

Fig. 2.3 shows the results of an immunodiffusion test with known antigens **P** and **Q** and the antibodies to these antigens.

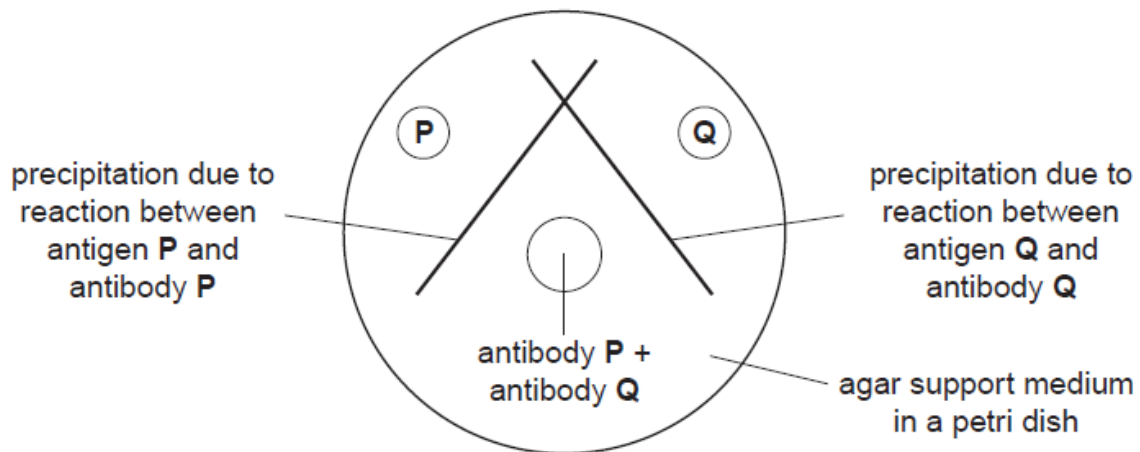


Fig. 2.3

In an investigation, the serum from two test organisms was tested for the presence of antibodies to specific antigens. Both organisms had been previously exposed to both antigens. The serum was placed in wells at the edge of the petri dish and the antigens in a central well.

Fig. 2.4 shows the test set-up.

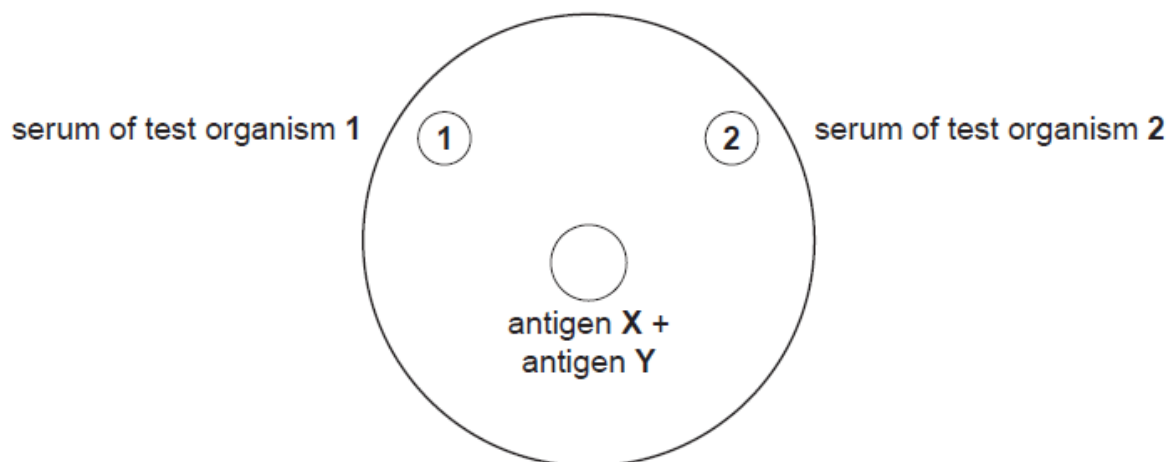


Fig. 2.4

(i) Suggest **one** variable that must be controlled in this procedure.

.....[1]

- 1 ref. to *any one*:
 thickness of agar
 / volume of agar
 / depth of wells
 consistency / concentration of, agar
 volume / diameter of wells
 distance of antigen wells from test organism wells / distance between wells
 temperature [*do not allow: pH*]
 volume of serum / antigen volume
 [REJECT: Concentration of serum/antibodies]

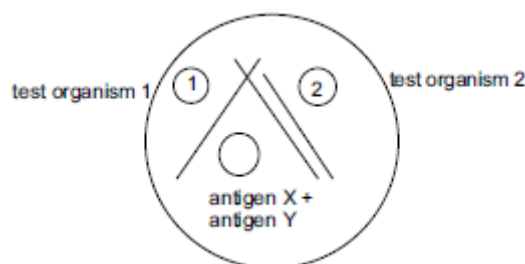
(ii) State the independent variable in this investigation.

..... [1]

- 1 the type of the **serum** / **antibody** / **antibodies** (from the test organism) ;
 [REJECT : antigen ; amount ; concentration ; types of test organisms – this is indirect independent variable]

(iii) Both test organisms had antibodies against antigen X, but only organism 2 had antibodies against antigen Y.

On Fig. 2.4 draw lines to represent where precipitation might have occurred for both organisms. [2]



- 1 Two lines / 1 thick line, between serum 2 and antigens ;
- 2 One line between 1 and antigens ;

[Additional guidance –

Allow: 2 marks if lines do not intersect

If the lines are reversed / spread outside dish max. 1

Do not allow: if the lines are inverted / lines cross wells]

(iv) Suggest one disadvantage of immunodiffusion for detecting antigens.

1. [1]

- 1 ref. to all antibodies not forming precipitates / AW ;
- 2 ref. to sensitivity / AW ; (*the idea that test will not detect low concentrations*)
- 3 ref. to more qualitative / difficult to quantify ;
- 4 ref. to, slow rate / inability to diffuse of some antibodies ; (*Allow – of 'slow' is in the context of getting results of tests*)
- 5 ref. to problem of identifying individual antigen [because an antibody e.g Ig M may potentially bind to several types of antigens] / AW ;

[REJECT :

Antigens mutate ;

Relating to experiment design issue - Difficult to tell which band of precipitate belongs to which set of reaction between antigen and antibodies. Because when using immunodiffusion for detecting antigen, one antibody is placed central well at a time / AW ;]

A **naturally occurring** mutant of *Plasmodium* sp. has been tested for use as a 'whole organism' vaccination against malaria. The mutant organism develops normally in mosquito vectors and infects the salivary glands in the same way as non-mutant wild type *Plasmodium* sp. In mice, the mutant infects liver cells but does not multiply and cannot enter red blood cells.

Trials using mice were carried out and the effectiveness of the mutant organism as a vaccine tested by injecting non-mutant wild type *Plasmodium* sp. into vaccinated and non-vaccinated mice.

Table 2.1 shows the results of investigations in mice using the mutant *Plasmodium* sp.

Table 2.1

test group	number of mutant <i>Plasmodium</i> cells given to the mice			percentage of mice not infected by wild type <i>Plasmodium</i> sp.
	first inoculation	first booster inoculation	second booster inoculation	
1	0	0	0	0
2	50 000	25 000	25 000	100
3	10 000	10 000	10 000	100
4	10 000	10 000	0	70

(v) Suggest the purpose of including each of the following test groups.

group 1

.....

groups 2 and 3

.....

group 4

.....[3]

- 1 *group 1*: ref. to the idea of a control ; to show that percentage change of mice not infected is due to inoculation ;
- 2 *group 2 & 3*: ref. to idea of finding how many organisms, give immunity / needed ; (*e.g. how many are needed to make the vaccine work*)
[REJECT: to determine or justify the minimum or maximum number of organisms needed]
- 3 *group 4*: ref. to idea of finding the number of, inoculations / boosters, needed ;
e.g. to see if another booster is needed

(vi) Using the information in the question, outline a procedure that might be used to obtain mutant *Plasmodium* sp. to use in the vaccination trials.

.....[2]

Salivary gland:

ref. to (information that mutant) Plasmodium breeds / develops / AW in mosquitoes ;

- 1 ref. to breeding mosquitoes / culturing in salivary gland tissue / AW ;
- 2 ref. to extracting (Plasmodium) from salivary glands / culture of cells from salivary glands ;

Liver:

ref. to (information that mutant) Plasmodium develops / AW in mice's liver cells ;

- 1 ref. to breeding infected mice / culturing (Plasmodium) in liver tissue / AW ;
- 2 ref. to extracting (Plasmodium) from mice's infected

[TOTAL : 19]

QUESTION 3: PLANNING QUESTION

Effect of citrate on rate of respiration

Enzymes catalysing essentially irreversible reactions are potential sites of control in cellular respiration. One of these enzymes is phosphofructokinase, which can be regulated by the reversible binding of citrate to its allosteric site. (Citrate is produced as an intermediate compound during Krebs cycle.)

Using this information and your own knowledge, design an experiment to determine the effect of citrate concentration on the rate of cellular respiration.

You must use:

- 10 mM citrate,
- purified homogenate of enzymes found in the cytosol,
- 5% glucose solution,
- pH buffer,
- distilled water,
- benedict's solution,
- apparatus shown in Fig. 3.1, can be used to separate proteins from ions and disaccharides.

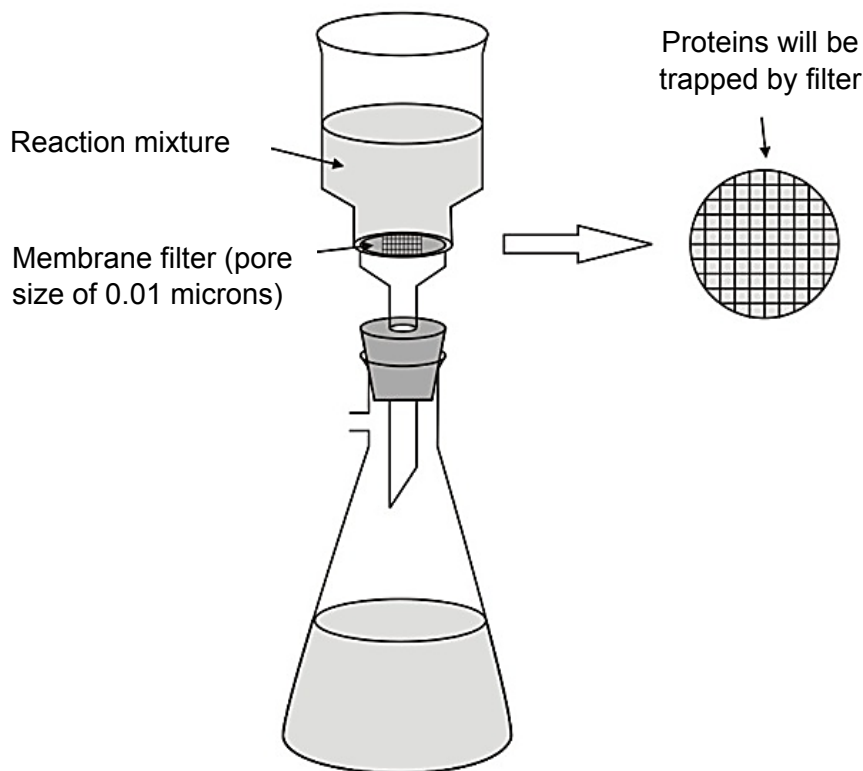


Fig. 3.1

- syringes,
- white card,
- stopwatch,
- thermometer,
- bunsen burner with tripod, gauze and bench mat,
- thermostatically controlled water bath,
- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 15]

MARK SCHEME

INTRODUCTION

A: BACKGROUND KNOWLEDGE / RATIONALE

[1m for 2 out of 3 points]

1. PFK is an enzyme involved in glycolysis where glucose is converted into pyruvate.
2. The rate of glucose consumption decreases as the concentration of citrate increases, since citrate acts as an inhibitor to phosphofructokinase.
3. Therefore the rate of respiration decreases.

[1m for 2 out of 3 points]

4. Presence of glucose can be detected using the Benedict's test.
5. The colour of the mixture / precipitate reflects the amount / quantity of glucose present.
6. As the concentration of glucose increases, the colour of the mixture / precipitate changes from blue to green to yellow to brick-red (any two colour)

[Rationale] – 1m for #2,3

1. **[Award marks for this under dependent variable section]** In this experiment, the effect of citrate on the rate of cellular respiration is determined by the rate of glucose consumption / amount of glucose consumed per unit time after a fixed period of time.
2. The concentration of glucose **remaining** can be estimated by comparing the results of the experiment against the results of the Benedict's test performed on a range of glucose solutions of known concentration / glucose standard solutions.
3. Concentration of glucose **consumed** can be calculated by subtracting the concentration of glucose remaining from the original 5%.

[Hypothesis] – 1m

1. As the citrate concentration increases, the rate of cellular respiration decreases as measured by the rate of glucose consumption.

B: VARIABLES AND CONTROLLED VARIABLES

[State the independent and dependent variables] – 1m for #1,2

1. The independent variable is citrate concentration / mM ; 0 mM, 1.5 mM, 3.0 mM, 4.5 mM, 6.0 mM, 7.5 mM, 9.0 mM. [At least 5 readings, regular intervals; maximum 10 mM]
2. The dependent variable is **rate** of respiration, calculated by concentration (in percentage) of glucose consumed per unit time

**[Other variables to keep constant: Can be written in detail in Procedure Section]
– 1 m for every 2 points; total 2m**

1. Volume of glucose

- Use a syringe to add the **same volume** (state volume here or in procedure) of fresh glucose solution from the same stock (stirred well before use) to keep the initial concentration of glucose constant.
2. Temperature
 - The optimum temperature, **e.g. 37°C**, is kept constant by using a thermostatically controlled water bath.
 3. pH
 - Keep pH constant with a pH buffer of same volume (2cm³)
 4. Duration of reactions
 - A digital stopwatch is used to ensure duration of reactions is kept constant **e.g. 3min.**

[Control] -1m

1. A control is set up with **2.0 cm³** of 0 mM citrate / **2.0 cm³** distilled water instead of citrate to show any change in colour of precipitate obtained from Benedict's test is due to presence of citrate.

C: DETAILED PROCEDURE – total 7m**Part 1: Preparation of the glucose standards**

1. Label 10 boiling tubes **0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5% and 5.0%**. (minimum 5 tubes)
2. Prepare 20.0 cm³ of various concentrations of glucose solutions as shown in the table below. 10.0 cm³ syringes are used to add the liquids and glucose solution is placed in their respective boiling tubes.

1m

Concentration of glucose solution to be prepared /%	Volume of 5% glucose solution / cm ³	Volume of distilled water / cm ³
0.5	2.0	18.0
1.0	4.0	16.0
1.5	6.0	14.0
2.0	8.0	12.0
2.5	10.0	10.0
3.0	12.0	8.0
3.5	14.0	6.0
4.0	16.0	4.0
4.5	18.0	2.0

3. Label 10 test-tubes 0.5%, 1.0%, 1.5%, 2.0%, 2.5%, 3.0%, 3.5%, 4.0%, 4.5% and 5.0%.
4. Using a 5.0 cm³ syringe, add 2.0 cm³ of each concentration of glucose solution into their respective test-tubes.
5. Using a 5.0 cm³ syringe, add 2.0 cm³ of Benedict's solution. Shake gently to mix the contents of the tube.
6. Place the test-tubes in the boiling water for two minutes. Start the stopwatch.
7. After two minutes, stop the stopwatch. Remove the tubes from the boiling water and place them in a rack.
8. Shake gently to mix the contents of the tube and observe the contents of the test-tubes immediately after mixing. Record the observations in a table, noting any differences in terms of colour and cloudiness.
9. Set aside these tubes for comparison in Part 3.

Part 2: Preparation of citrate solution

1. Label 7 boiling tubes 0 mM, 1.5 mM, 3.0 mM, 4.5 mM, 6.0 mM, 7.5 mM, 9.0 mM.
[At least 5 readings, regular intervals; maximum 10 mM]
2. Prepare 10.0 cm³ of various concentrations of citrate solution as shown in the table below. 10.0 cm³ syringes are used to dispense the liquids and citrate solution into their respective boiling tube.

Concentration of citrate /mM	Volume of 10 mM citrate /cm ³	Volume of distilled water /cm ³
1.5	1.5	8.5
3.0	3.0	7.0
4.5	4.5	5.5
6.0	6.0	4.0
7.5	7.5	2.5
9.0	9.0	1.0

1m

Part 3:

3. Label another 7 boiling-tubes 0 mM, 1.5 mM, 3.0 mM, 4.5 mM, 6.0 mM, 7.5 mM, 9.0 mM.
4. Using two clean 5.0 cm³ syringes, add 5.0 cm³ of 5% glucose solution [Note: **concentration of glucose provided**] and 2.0 cm³ of pH buffer (e.g. 6.2) into a boiling-tube labeled 1.5 mM.
5. Place this boiling tube and the purified homogenate of enzymes into a thermostatically controlled water bath at 37°C for 10 minutes to **equilibrate**. Start the stopwatch.
6. Also **equilibrate** the **citrate** solutions in a thermostatically controlled water bath at 37°C for at least 10 minutes.
7. After 10 minutes, stop the stopwatch. Using a 2.0 cm³ syringe, add 2.0 cm³ of 1.5 mM citrate solution and using a 5.0 cm³ syringe, add 5.0 cm³ of enzyme homogenate into the boiling-tube. [Note: **enzyme homogenate should be the last to be added.**] Shake gently to mix the contents of the tube.
8. Place the test tube into a thermostatically controlled water bath of 37°C for 3 minutes. Start the stopwatch.
9. After 3 minutes, quench/stop the enzymatic reaction by placing the test tube in the boiling water bath for 2 minutes.
10. After 2 minutes, stop the stopwatch and pour the reaction mixture through the filtration set-up / membrane filter.
11. Perform Benedict's test on the filtrate as it was done for the glucose standards (steps 4-8, Part 1).

1m

12. Place the tubes containing the filtrate and the glucose standards against a white card. Compare with the glucose standards (from Part I, Step 9) to estimate the concentration of glucose present. Shake the tube gently to mix the contents before comparison.
13. Record the concentration of glucose remaining /% in a table.
14. Calculate the concentration of glucose consumed (5% - concentration of glucose remaining) / %
15. Repeat steps 4 to 14 using the other citrate concentrations as prepared in Part 2.

[Replicates and Repeats] – 1m

16. To ensure reliability of results, repeat steps 4 to 15 to obtain a total of three readings (triplicates) at each citrate concentration, and calculate the average.
17. To ensure reproducibility of data, repeat the entire experiment twice using freshly prepared reagents and solutions and glucose standards.

D: DATA MANIPULATION AND EXPECTED RESULTS**1. [Draw Table of results] -1m**

Table showing the amount of glucose /% at different concentrations of citrate / mM

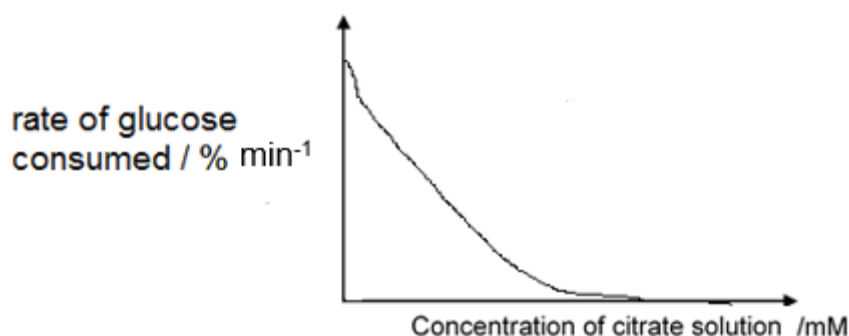
Concentration of citrate /mM	Concentration of glucose remaining/ %				Concentration of glucose consumed / %	Rate of respiration / % min ⁻¹
	Reading 1	Reading 2	Reading 3	Ave		
0.0						
1.5						
3.0						
4.5						
6.0						
7.5						
9.0						

Concentration of glucose consumed = 5% - Concentration of glucose remaining

Rate of respiration = Concentration of glucose consumed / 3 min

2. [Draw a graph to show expected trends and/or results] -1m

Graph of rate of glucose consumed /% min⁻¹ against concentration of citrate /mM

**SAFETY PRECAUTIONS -1m for 2 sets**

1. Wear safety goggles and gloves when handling chemical reagents which may be irritants (e.g. Benedict's solution, citrate solution) to the eyes and/or skin. Immediately flush the eye / wash with water when the chemical reagent comes into contact with the eyes and/or skin. If irritation persists, call for medical help.
2. Handle glassware with care as broken glassware is sharp and can cause cuts.
3. AVP

[Total: 15]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

PREPARATION LIST FOR QUESTIONS 1 AND 2

S/N	Apparatus/Reagents/Chemicals	Quantity per student
1	In a beaker, labelled S , 1.0% starch solution	at least 40 cm ³
2	In a beaker, labelled E , 5% amylase solution	at least 40 cm ³
3	In a beaker, labelled W , distilled water	at least 100 cm ³
4	In a dark bottle, labelled iodine , iodine (in potassium iodide) solution- used for testing for the presence of starch	at least 20 cm ³
5	Test-tube rack, suitable to hold 7 test-tubes	1
6	Test-tubes	7
7	Beakers or containers to hold up to 50 cm ³ / plastic containers	6
8	Beaker to contain about 400 cm ³ of water, labelled water-bath	1
9	Access to a source of cold (tap) water to prepare a water-bath at about 35 °C	
10	Access to a source of hot water at 50 °C or more to prepare a water-bath at about 35 °C	
11	250 cm ³ beakers	2
12	Thermometer, –10 to 110 °C	1
13	Paper towels	8
14	10 cm ³ syringes	2
15	5 cm ³ syringes	2
16	Pasteur pipette, 1 cm ³ or 3 cm ³	2
17	White tiles (10 cm × 10 cm)	1
18	Glass rod	1
19	Stopwatch	1
20	Glass marker pen	1
21	Safety glasses/goggles	1 pair
22	Coloured photo micrograph of Fig. 2.1	1
23	Light microscope	1 per pair of candidates
24	Plastic ruler	1

INSTRUCTIONS FOR PREPARING APPARATUS FOR QUESTION 2

These instructions give details of the apparatus required by each candidate for each experiment in this paper. A summary of the questions that will be presented to the candidates is included, where appropriate, to allow the biology teacher to test the apparatus appropriately. **No access to the Question Paper is permitted in advance of the examination.**

Candidates must be provided with a microscope with:

- Eyepiece lens, $\times 10$ (equal to 16 mm or 23")
- Low-power objective lens, $\times 10$ (equal to 16 mm or 23")
- High-power objective lens, $\times 40$ (equal to 4 mm or 16")
- Eyepiece graticule fitted within the eyepiece and visible in focus at the same time as the specimen.

To avoid confusion, only the lenses specified above should be fitted in the microscopes to be used in the examination. Any lenses which are **not** $\times 10$ or $\times 40$ should be removed or replaced. Each candidate must have sole, uninterrupted, use of the microscope for at least one hour.

Each candidate will require:

- (i) Clear plastic ruler, marked in mm
- (ii) Microscope (as described above)

For each candidate:

- the microscope **must** be set up on low power
- the slide must **not** be left on the stage of the microscope.

APPARATUS LIST FOR QUESTIONS 1 and 2

S/N	Apparatus/Reagents/Chemicals	Quantity per student
1	In a beaker, labelled S , 1.0% starch solution	at least 40 cm ³
2	In a beaker, labelled E , 5% amylase solution	at least 40 cm ³
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5	Test-tube rack, suitable to hold 7 test-tubes	1
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8	Beaker to contain about 400 cm ³ of water, labelled water-bath	1
9	Access to a source of cold (tap) water to prepare a water-bath at about 35 °C	
10	Access to a source of hot water at 50 °C or more to prepare a water-bath at about 35 °C	
11	250 cm ³ beakers	2
12	Thermometer, –10 to 110 °C	1
13	Paper towels	8
14	10 cm ³ syringes	2
15	5 cm ³ syringes	2
16	Pasteur pipette, 1 cm ³ or 3 cm ³	2
17	White tiles (10 cm × 10 cm)	1
18	Glass rod	1
19	Stopwatch	1
20	Glass marker pen	1
21	Safety glasses/goggles	1 pair
22	Coloured photo micrograph of Fig. 2.1	1
23	Light microscope	1 per pair of candidates
24	Plastic ruler	1